

Development of Machine Learning-Based Surrogate Models for Dynamic Chemical Processes

Group: Project Group 1

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Abstract:

Our project aims to create a surrogate model of a dynamic chemical process model using machine learning algorithms. The advantage of this lies in the faster solving time of the ML model which plays an important role in online applications in the process industry, e.g. in model predictive control.

In the first step we will create a dynamic process model of a simple chemical process like a CSTR with a reaction or a distillation column.

In the second step we will then simulate the model to create data which we can then use for training the surrogate model. Finally, the success of our project will be evaluated according to multiple criteria. The surrogate model will be compared to the first-principles model based on both solving time and accuracy.

Since the systems we model are dynamic, the majority of the data we work with will be time series. The goal of our machine learning model will therefore be to be able to predict subsequent values based on previous ones, not simply input-output pairs, so we will use an autoregressive model to achieve this. According to the paper [1], NARX and NOE Models have different strengths regarding predication and simulation. We would like to verify these findings in our project.

[1] Rebello, C. M., Marrocos, P. H., Costa, E. A., Santana, V. V., Rodrigues, A. E., Ribeiro, A. M., & Nogueira, I. B. (2022). Machine learning-based dynamic modeling for process engineering applications: a guideline for simulation and prediction from perceptron to deep learning. *Processes*, 10(2), 250.